# Igor Bryzhikov

*Curriculum Vitae*

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## Education

2025 – 2029 (anticipated) **Moscow Institute of Physics and Technology**, *Phystech School of Applied Mathematics and Computer Science*, B.Sc. in Informatics and Computer Engineering

**Relevant coursework:**

Mathematics: Calculus, Multivariable Calculus, Analytic Geometry, Linear Algebra, Differential Equations

Computer Science: Algorithms and Data Structures, C++, Python (advanced course), Software Engineering, Databases and SQL

Physics: Mechanics, Thermodynamics and Molecular Physics, Electricity and Magnetism, Analytical Mechanics

**Average Grade:** 8.7/10

## Research Experience

Oct. 2026 – present **Research project**, *MIPT*, **Supervisor:** Prof. Alexander V. Gasnikov

Paper in preparation: *"Algorithmic $C^p$-dimensional Lower Bounds: Deterministic Noise in Zeroth-Order Convex Optimization"*. Studying the oracle complexity of gradient-free (zeroth-order) convex optimization when function values are corrupted by deterministic, non-random noise; constructing lower bounds for $C^p$-smooth objectives and comparing them with known upper bounds for finite-difference and smoothing-based methods.

## Work Experience

Jul. 2026 – Oct. 2026 **BI Analyst Intern**, *T-Bank*, Data Security team

Automated the processes of a team that detects and masks sensitive data in the bank's data warehouse.

- Designed and calculated an efficiency metric for automated alert routing to data owners, showing a 24% reduction in operational load; re-tuned table re-scan prioritisation to meet the quarterly target.
- Built two LLM tools: automated pre-review of Jira tasks (mentor review load down 2.5×) and a hint on masked fields for access requests (time to request down from 9 to 2.75 min).

## Projects

Mar. 2026 – present **Training-data pipeline for LLMs**, *Data2Use startup*

Designed the pipeline stages for building LLM training datasets from unstructured corporate data; implemented validation, preprocessing and data-quality analytics (duplicates, gaps, schema errors) with unit tests; cut masking errors by 75%.

Nov. 2025 – present **Conversational AI companion with a user profile**, *personal project*

Designed the database schema for model state and the rules for updating it; implemented the backend (Python, SQLite, SQLAlchemy, REST API) with background jobs; covered 78% of critical endpoints with tests and reduced mean response latency from 340 to 190 ms.

Feb. 2025 – Apr. 2025 **Real-time threat detection in video streams**, *Russian National Entrepreneurship Olympiad, final*

Set recognition quality requirements, fine-tuned YOLOv8 CV-model and built an application that processes a live video stream (Python, PyTorch).

## Honors & Awards

2025 **Final of the Russian National Entrepreneurship Olympiad** (technology track), Finalist

**Interregional Mathematics Olympiad for School Students** (departmental universities), Prize-winner

## Extracurricular Experience

Jul. 2026 – present **Teacher gratitude platform** (MIPT and Phystech Union): responsible for outreach to former MIPT faculty.

Oct. 2025 – present **"Generation" mentorship programme**: group mentored by Viktor Kuznetsov, founder of a company with annual revenue of over RUB 180 billion (VseInstrumenti.ru).

## Computer Skills

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|---|---|
| Python | Jupyter, Data Analysis and Visualization (NumPy, Pandas, MatPlotLib), PyTorch, FastAPI |
| ML | LLM, RAG, LoRA fine-tuning, prompt engineering, model evaluation, ML-Infra |
| C++ | Object-Oriented Programming, Algorithms and Data Structures, Metaprogramming |
| SQL | Analytical SQL (CTEs, window functions, analytical queries, joins), query optimisation, SQLAlchemy |
| Other | Linux, Git, Bash, Docker, CI/CD, Apache Superset |

## Languages

**Russian** (Native), **English** (B2 – Upper-Intermediate)